

Mathematics - Class 10

Description: Sample mathematics book for testing PDF upload in AI Knowledge Bot.

Chapter 1: Real Numbers

Real numbers include rational and irrational numbers. Euclid's division lemma helps find the HCF of two numbers.

Chapter 2: Polynomials

A polynomial is an algebraic expression consisting of variables and coefficients. The degree of a polynomial is the highest power of the variable.

Chapter 3: Linear Equations

A pair of linear equations in two variables can be solved using graphical, substitution, or elimination methods.

Chapter 4: Quadratic Equations

A quadratic equation has the form $ax^2 + bx + c = 0$ where $a \neq 0$. It can be solved using factorization, completing the square, or the quadratic formula.

Chapter 5: Trigonometry

Important identities: $\sin^2\theta + \cos^2\theta = 1$, $\tan\theta = \sin\theta/\cos\theta$. Applications include finding heights and distances.

Practice Questions

1. Find the HCF of 96 and 144.
2. Solve $x^2 - 5x + 6 = 0$.
3. Solve: $2x + y = 7$ and $x - y = 2$.
4. Evaluate $\sin 30^\circ$ and $\cos 60^\circ$.